Which apartments are over-priced, and how sure are we? Location, photos, text and a trust layer on krisha.kz ads

Final: the Endterm model as the base, three new information sources fused and kept only if they beat it by the fixed 0.354 M rule, calibrated intervals for the user

Course project, Final (Week 10 defence) · Machine Learning Algorithms · Direction 1 (Apartments for sale) · Regression

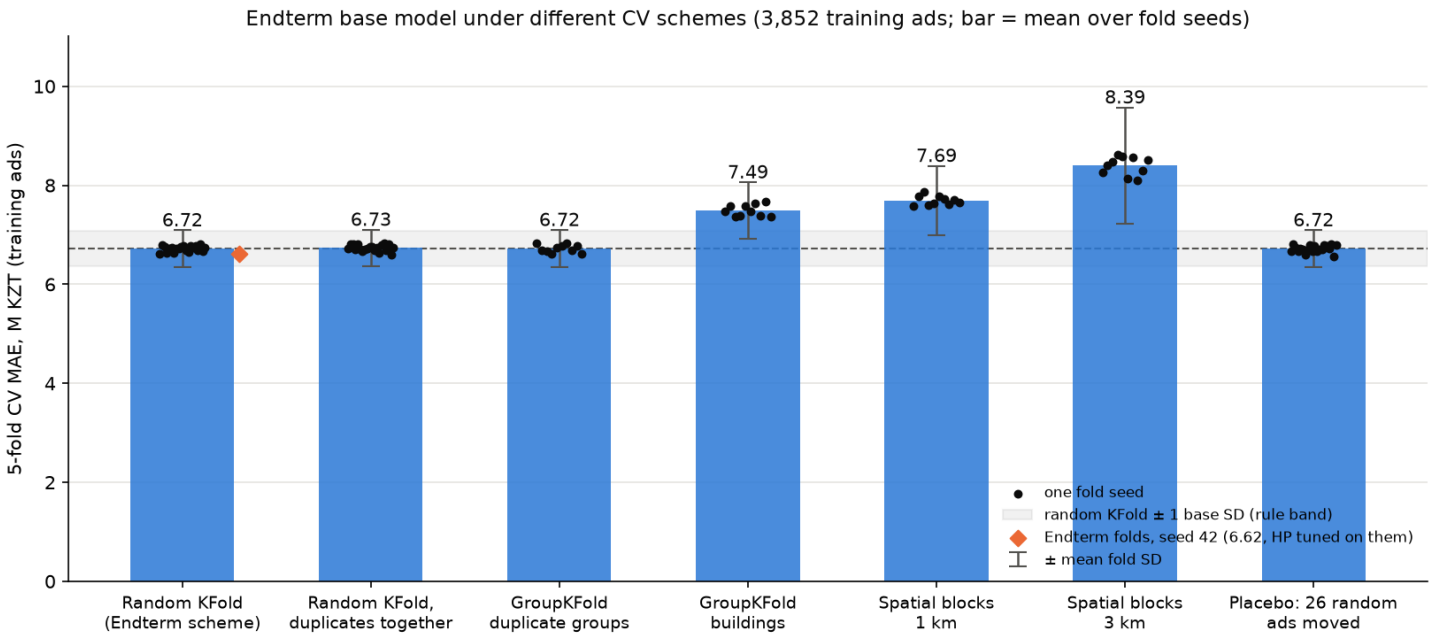
Rakhat Balgabekov, Aitbek Kaiyrbekuly · Group SE-2424 · November 2026

Question, base and what changed since the Endterm

Which apartments in Astana and Almaty are asked above or below what comparable apartments go for, by how much, and how sure can we be?

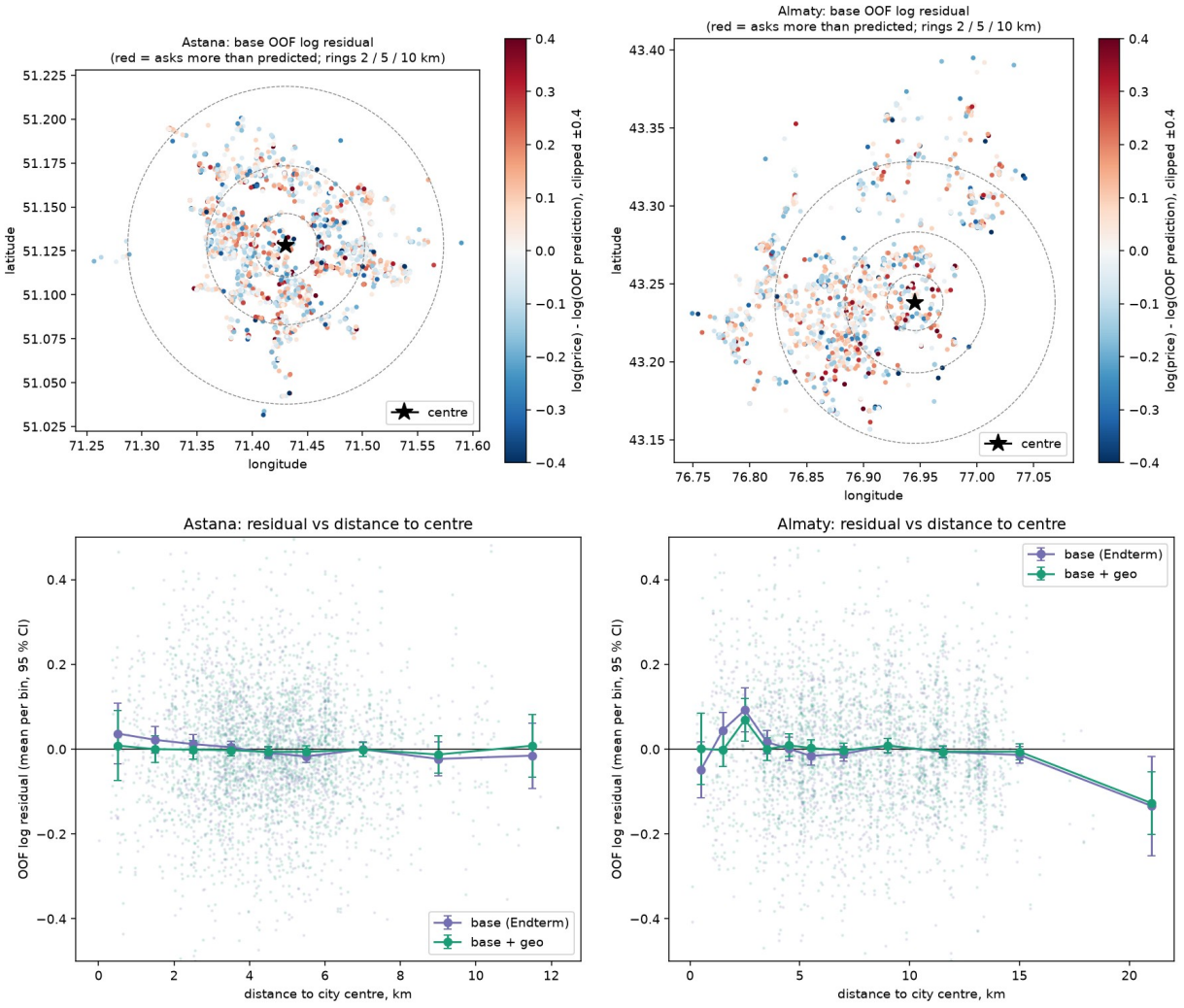
- **Same data and split** 4,815 ads (Oct 2026), 3,852 training ads, the 963 saved test ads used once
- **Base = Endterm model, reproduced exactly** tuned HGB on log price, CV 6.616 ± 0.353 M, test MAE 6.29 M, MdAPE 9.13 %
- **One rule for every model** kept only if CV MAE < base − 0.354 M (Endterm fold SD, fixed) = 6.370 M on duplicate-aware folds (6.262 M random)
- **New test** 1,507 November ads, a month later, search-page level only
- **Leakage audit** duplicate ads found by photo hashes, embeddings, fields and text; grouped CV
- **Location** OSM distances and counts (parks, schools, shops, centre)
- **Photos** CLIP zero-shot, linear probes, fine-tuned EfficientNet-B0, Grad-CAM, mismatch flag
- **Text** TF-IDF vs transformer embeddings as residual correctors; red flags
- **Trust layer** conformal 80/90 % intervals, SHAP, 5 comparable ads, interval-based over-pricing flag

Leakage audit: duplicates exist, but too few to bend the Endterm CV



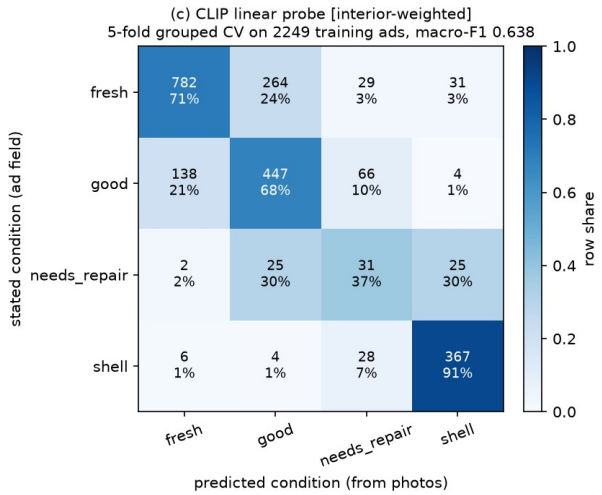
- **4 signals** pHash/dHash of photos, image-embedding cosine, structured fields (area 1 %, price 5 %, 150 m), text TF-IDF; stock photos (> 5 ads) ignored
- **Found** 48 pairs → 43 groups, 90 ads; 18 certainly the same flat
- **Test split** 13 test ads (1.35 %) have a training twin
- **Repaired folds** Endterm folds with each duplicate group in one fold: base 6.723 ± 0.264 M vs 6.616 M random; over 20 seeds duplicates add only +0.015 M
- **Decision** repaired folds = primary scheme for the model choice; margin kept at 0.354 M (fold SD moves 0.26–0.50 with the seed; the earlier 6.459 threshold was wrong)

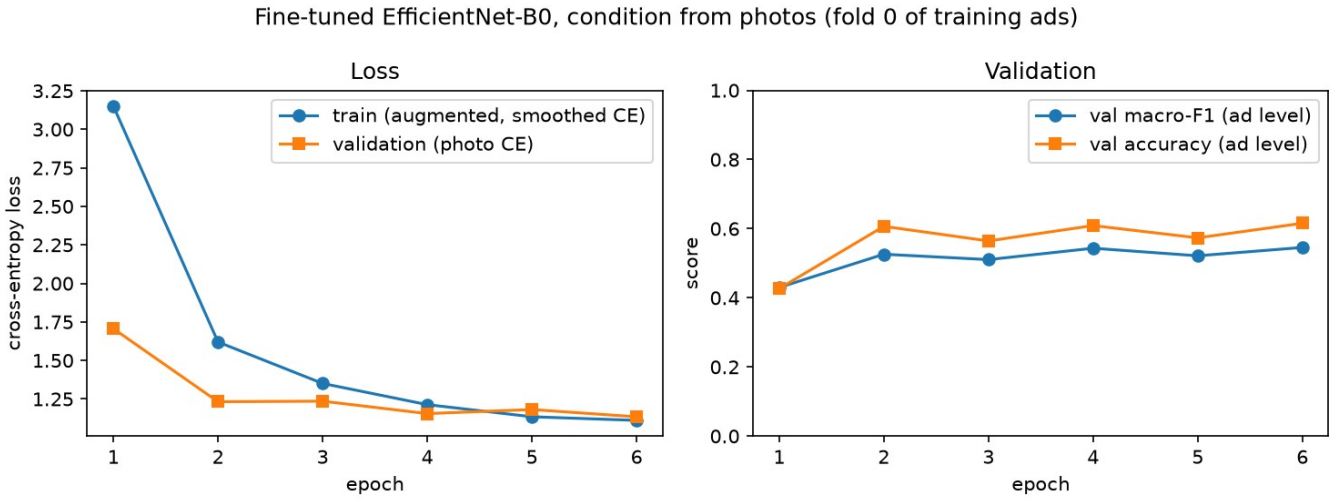
Location: a smooth distance to the centre helps; more OSM objects add little



- **Features** haversine distance to centre, nearest park / school / kindergarten / shop / clinic, counts within 500 m and 1 km (BallTree, OSM Overpass)
- **Best variant** base + geo + raw fields: 6.481 ± 0.280 M (-0.135, 4/5 folds better)
- **Distance to centre alone** 6.543 M, most of the OSM gain
- **Raw ad-page fields alone** 6.766 M: sparse columns add noise
- **1-SD rule** 0 of 4 variants pass

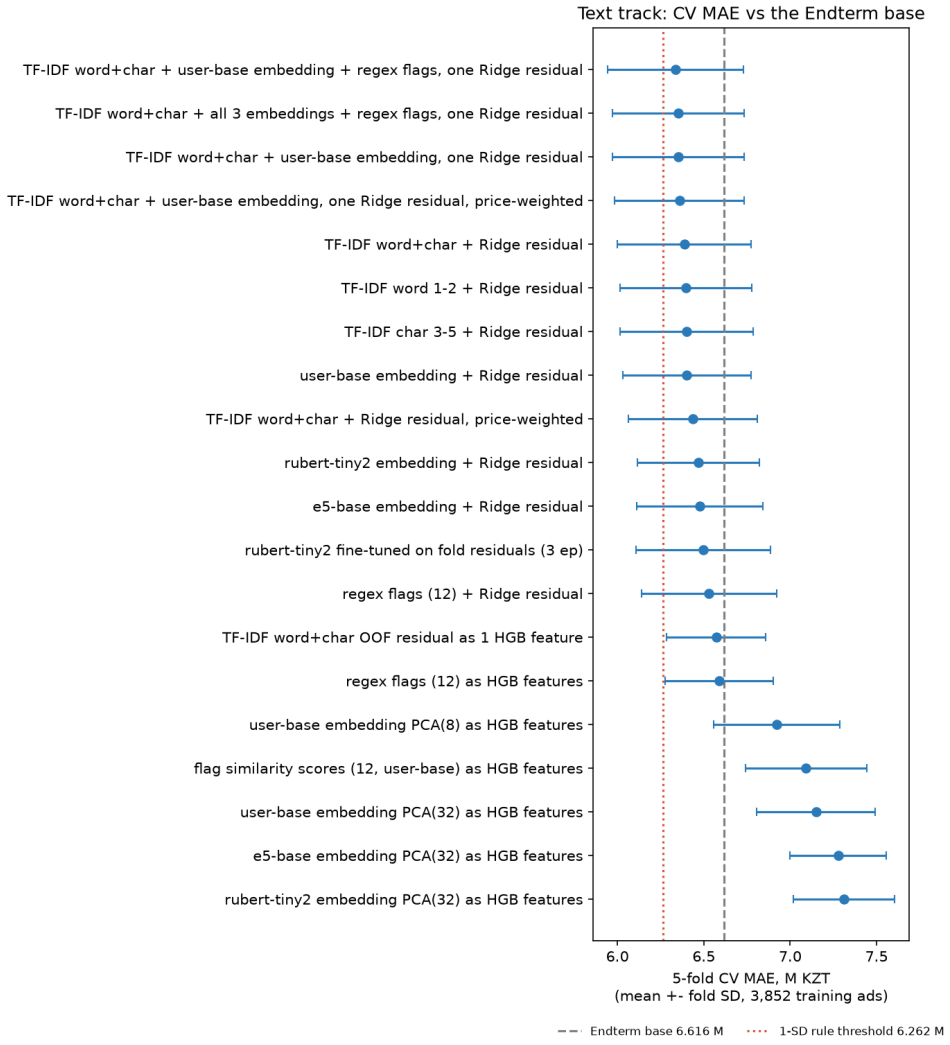
Photos: CLIP reads the renovation; the price model gains little from it





- **Task the photos can see** flat condition (fresh / good / needs repair / shell), grouped CV by duplicate photos
- **Frozen CNN + LR** MobileNetV3 F1 0.592, EfficientNet-B0 0.608
- **CLIP linear probe** macro-F1 0.632; fine-tuned EfficientNet-B0 0.546 (fold 0)
- **Grad-CAM** where the network looks; mismatch flag when photos contradict the stated condition
- **In the price model** best 6.426 M (-0.190); 0 of 14 pass

Text: a residual correction that helps on every fold, but below the bar



- **Sanitised first** money sentences, digits and contacts removed, so the text cannot leak the price
- **As a residual corrector** Ridge on the base model's inner-OOF log error, alpha by GCV per fold
- **Best** TF-IDF word+char + user-base embedding + regex flags, one Ridge residual: 6.336 ± 0.392 M (-0.280, 5/5 folds)
- **TF-IDF vs transformer** 6.388 vs 7.150 M, equal; embeddings as tree features 7.313 M, worse
- **1-SD rule** 0 of 20 pass; red flags shown to the user, too rare to move the MAE

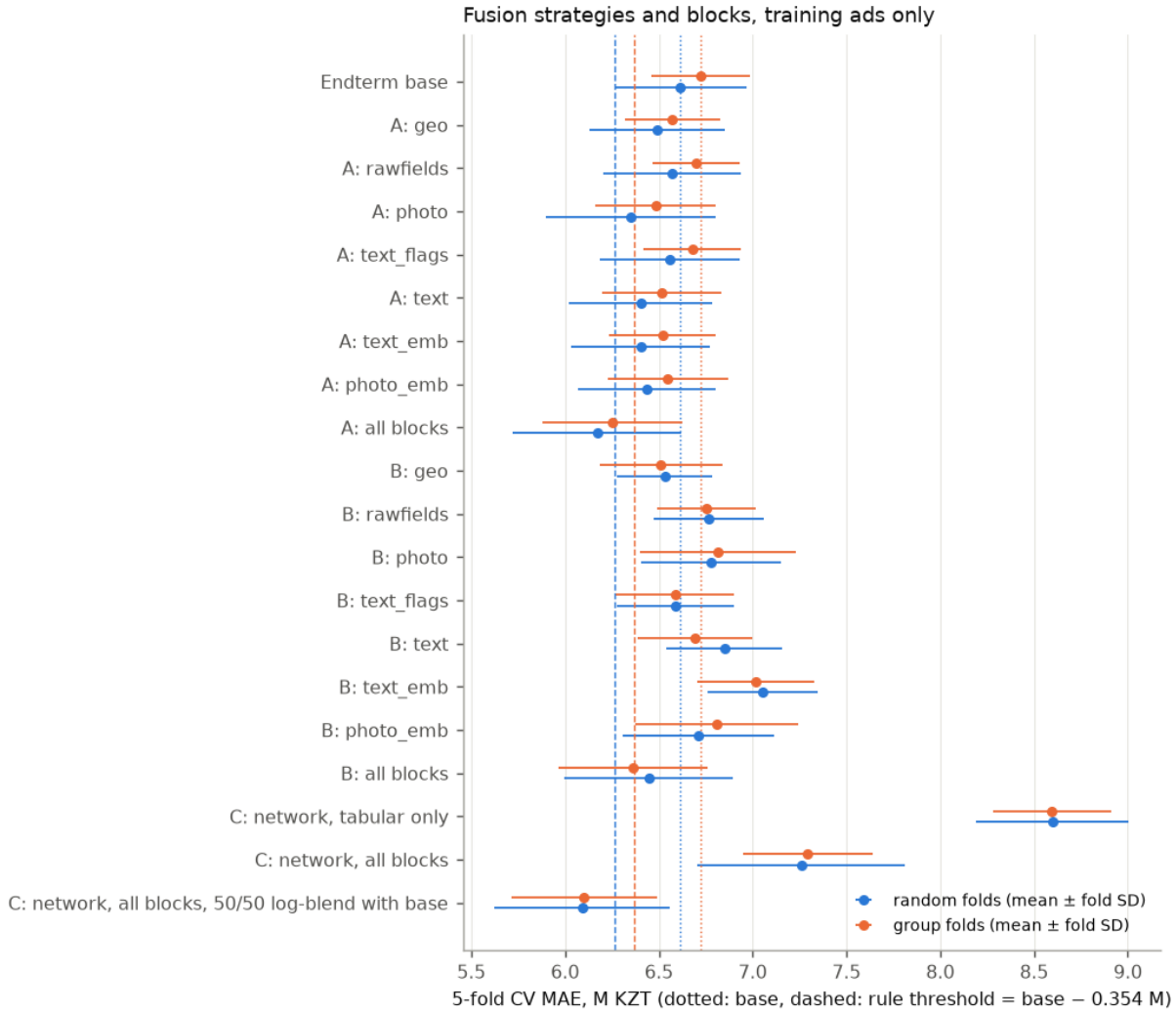
Fusion and selection: per-step vs whole model

block	best variant	CV MAE (M KZT)	Δ vs base (M)	folds better	variants kept
location (OSM + raw fields)	base + geo + raw fields	6.481 ± 0.280	-0.135	4/5	0/4
photos	residual: HGB on condition probs + zero-shot	6.426 ± 0.440	-0.190	4/5	0/14
text	TF-IDF word+char + user-base embedding + regex flags, one Ridge residual	6.336 ± 0.392	-0.280	5/5	0/20
fusion of kept blocks	C: network, all blocks, 50/50 log-blend with base	6.092 ± 0.468	-0.524	5/5	5/38

CV = 5-fold CV on the 3,852 training ads (mean ± fold SD). Rule: kept only if CV MAE < base − 0.354 M (6.370 M, repaired folds).

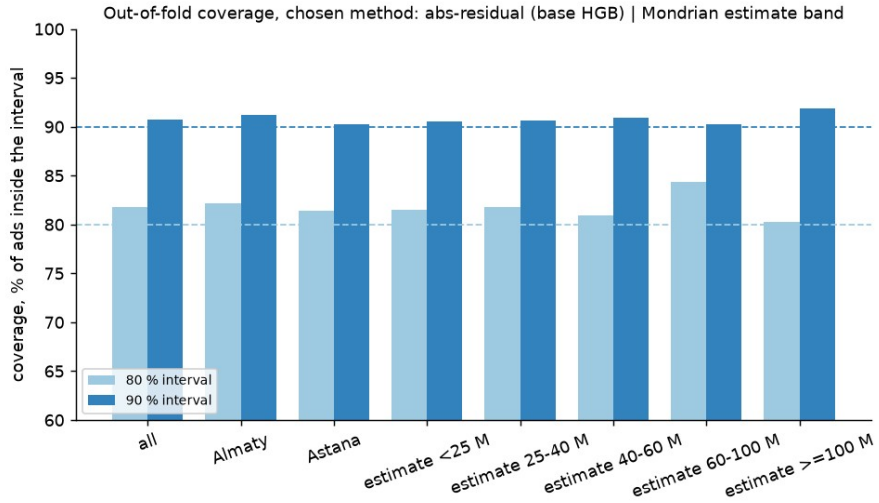
- **Per-step forward selection (strict)** no block passes; the largest first step (photo features, −0.24 M) < 0.354 M → would keep the base
- **Whole model** network + base 50/50 log-blend 6.100 ± 0.388 M, −0.623 M > 0.354, 5/5 folds
- **Not just ensembling** blend with a tabular-only network is worse than the base
- **Random folds agree** 6.092 M vs 6.616: not from duplicates
- **Final = the blend** certified as a whole, no single block certified

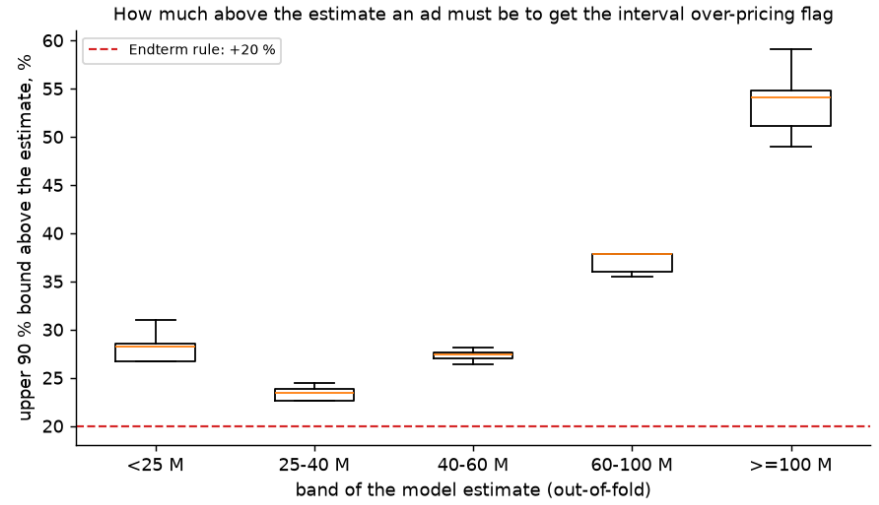
Fusion: every variant against the rule, on random and on repaired folds



- **A = one block added to the base** every single block lands between the base and the rule line
- **B = blocks as residual correctors** same picture, slightly worse
- **C = network** tabular-only network is worse than the base; with all blocks and the 50/50 blend it passes on both fold schemes
- **Reading** the gain comes from the blocks together, not from the blend trick

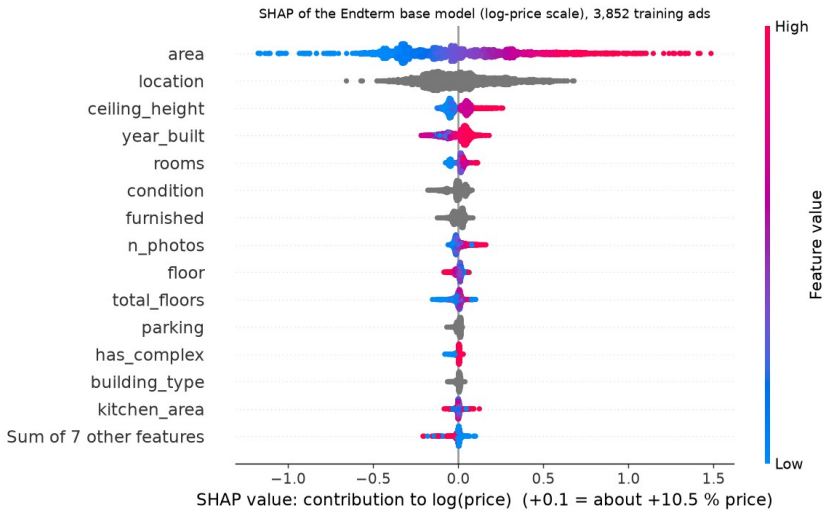
Conformal intervals replace the fixed 20 % over-pricing rule

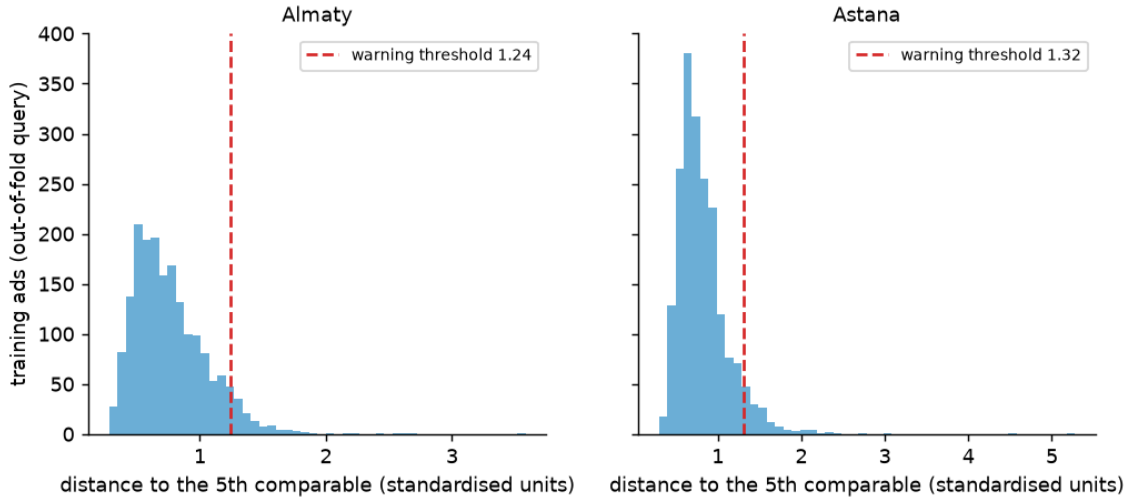




- **Conformal** out-of-fold scores, quantile $\lceil (n+1)(1-\alpha) \rceil / n \rightarrow$ guaranteed average coverage
- **Compared** constant width on log residual vs conformalized quantile regression (CQR), per-city calibration
- **Chosen** abs-residual (base HGB) | Mondrian estimate band: 90 % coverage 90.7%, median width 48 % of the price
- **Flag** over-priced if the price is above the upper 90 % bound; overlap with the 20 % rule 0.49 (Jaccard)
- **Why** 20 % is too strict for standard flats, too lenient for unique ones

Reasons and comparables: SHAP for the model, real ads for the user

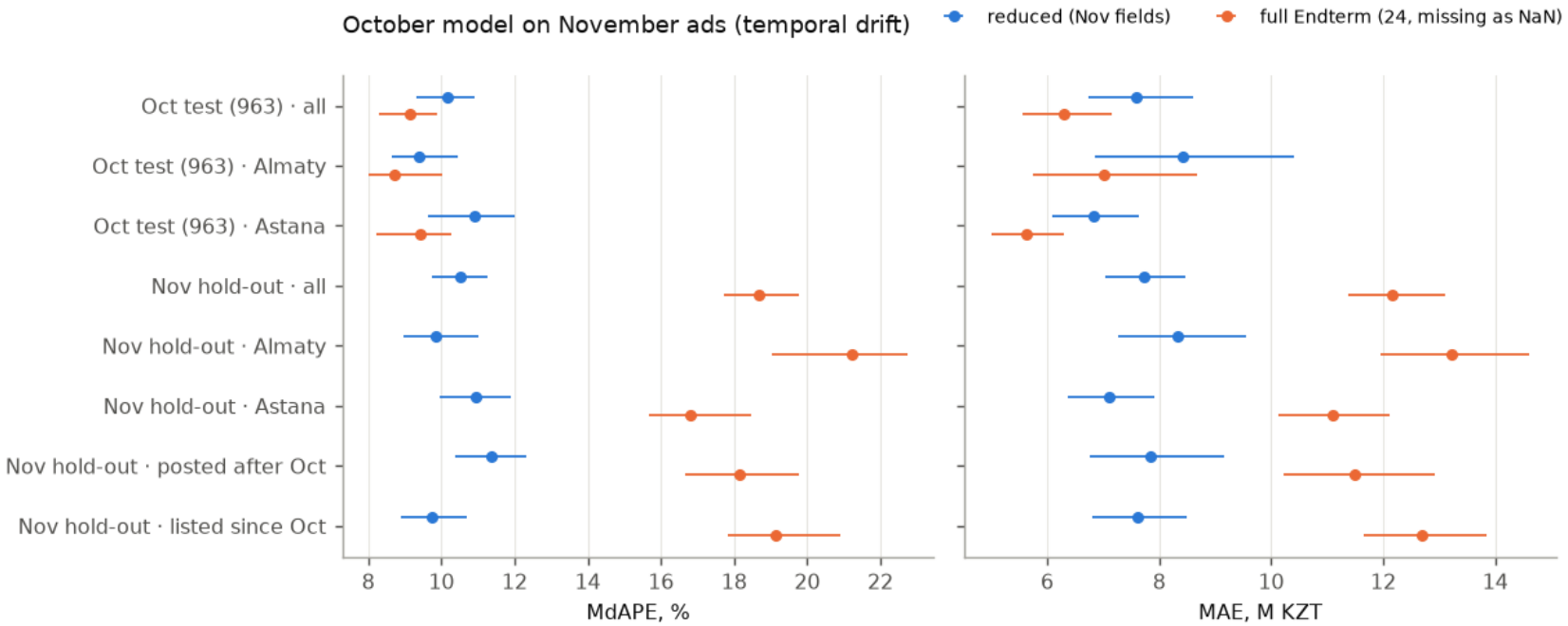




- **TreeSHAP** exact Shapley values, one-hot columns summed to the source feature
- **Top features** area, location, ceiling_height; stable vs ordinal encoding (ρ 1.00)
- **Comparables** 5 nearest training ads in the same city (area, rooms, year, floor, position)
- **Warning** 5th neighbour far → few comparables: 5.1% of ads, larger errors there
- **Limit** SHAP explains the model, not the market

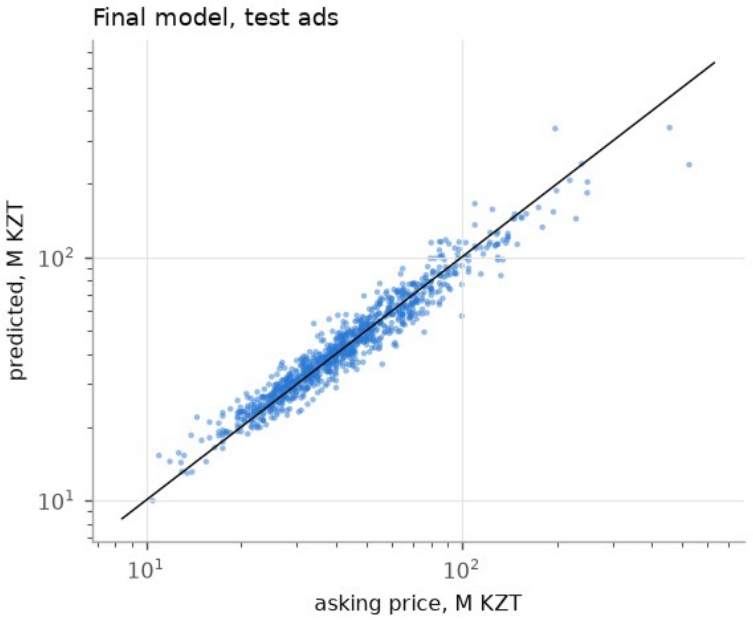
One test evaluation, and the same model a month later

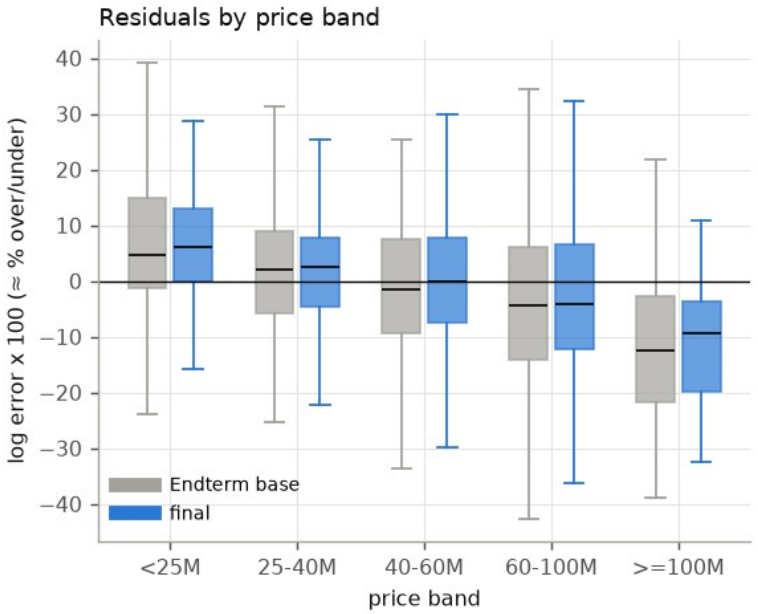
model	CV MAE, M	CV MdAPE, %	test MAE, M	test MdAPE, %	test R^2	within ±20 %
Endterm base (tuned HGB)	6.62	8.93	6.29	9.13	0.855	84%
Final: network + base, 50/50 log-blend	6.09		5.80	7.82	0.853	88%



- **963 saved test ads, once** MAE 5.80 M (95 % 5.09–6.67), MdAPE 7.82 %, ±20 %: 88%
- **vs Endterm, paired** MAE 6.29 → 5.80 M, difference -0.80 to -0.16 M; RMSE, R^2 unchanged (tails)
- **November (1,507 ads, card fields only)** reduced model MdAPE 10.16 % (Oct test) → 10.51 %: small drift
- **Read with care** test set reused (2nd use); Nov ad pages blocked by bot protection → fused model not testable there

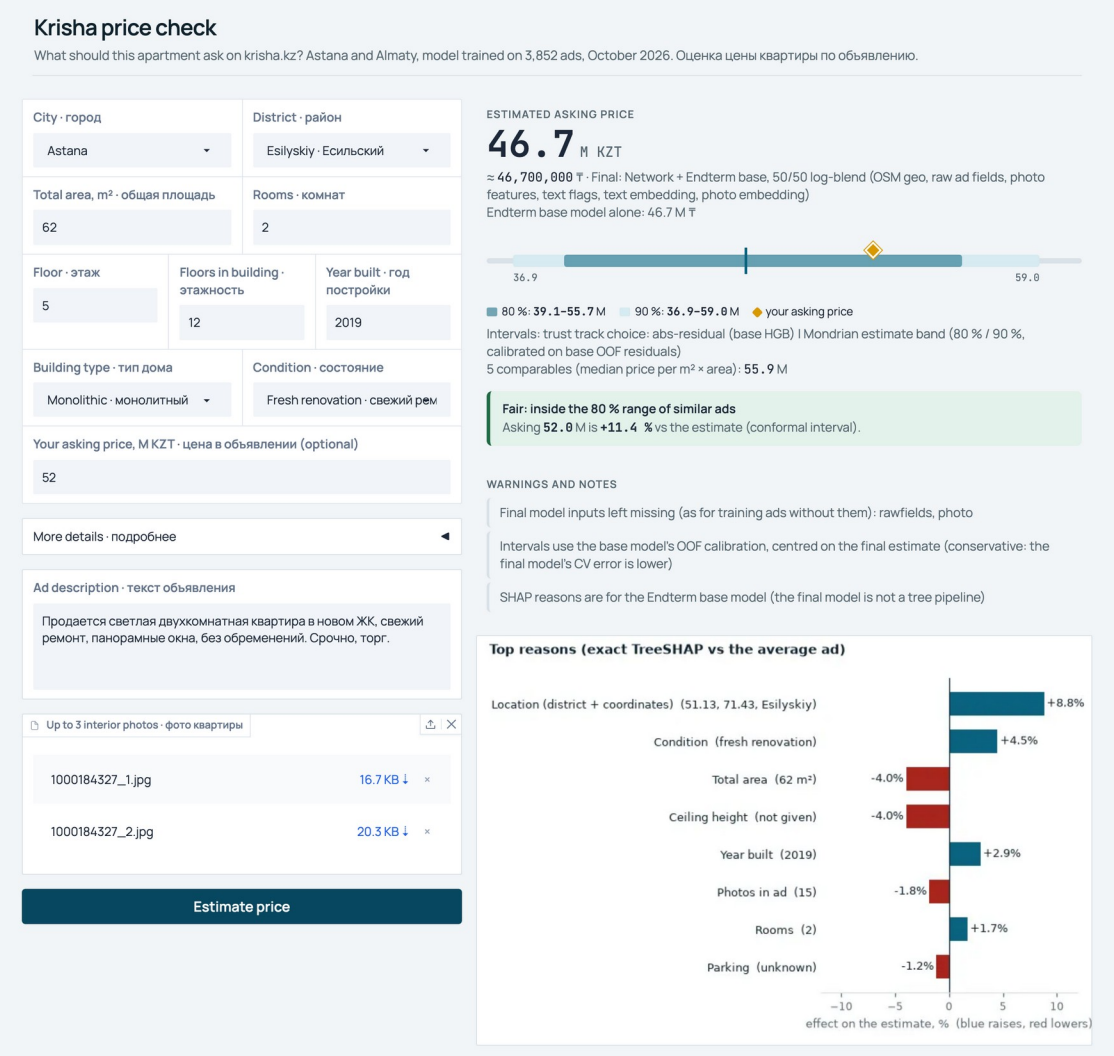
Where the final model is still wrong





- **Absolute error grows with price** MAE 2.1 M below 25 M, 28.9 M above 100 M
- **Shrinkage to the mean** top band under-predicted (−9 %), cheapest over-predicted (+6 %)
- **Gain is in the bulk** better in every city, band and room count except 4+ rooms (+0.84 M, CI crosses 0)
- **Worst 20 errors** 17 are flats above 100 M, where the intervals are widest

The app: one listing end to end



- **Form → final model** geo from coordinates, text flags + USER-base embedding from the description, CLIP embedding of up to 3 photos
- **Missing by design** raw ad-page fields and photo-track table features (as for training ads without them); the app says so in its notes
- **Around the estimate** conformal 80/90 % interval and verdict, SHAP reasons, 5 comparables, red flags, photo condition + mismatch check
- **Invalid input** reported field by field, nothing crashes
- **Run** python app/gradio_app.py → http://127.0.0.1:7860

Why no deep RL, and limitations

- **No deep RL** RL needs actions and rewards (sold? when? at what price?); we have asking prices only, no outcome, no days-on-market, one snapshot → it would collapse into our regression with more variance
- **Right tool** supervised regression + calibrated uncertainty; RL only with longitudinal data
- **Asking ≠ sale price** 'over-priced' = above comparable asking prices
- **Reused test set** 963 ads scored at the Endterm and once more here
- **November = listing level only** ad pages blocked by bot protection; 17 of 24 fields
- **Whole-model certification only** no single block passes step by step; removing location costs the blend most (+0.29 M)
- **Also** one snapshot, seller-stated labels, OSM gaps, 5 folds a coarse ruler

Contributions and AI tools

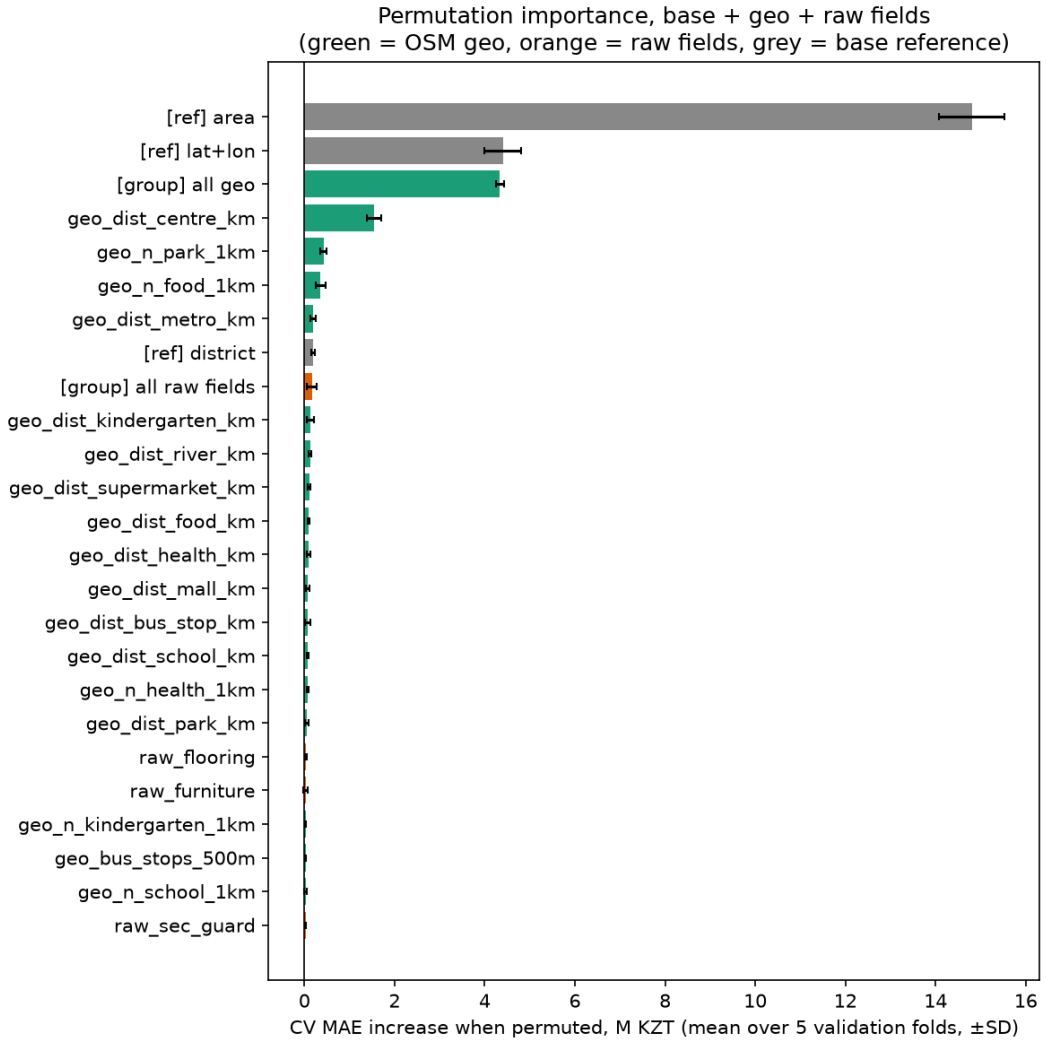
- **Rakhat Balgabekov** krisha_ml foundation, Endterm reproduction, leakage audit and repaired folds, photo track, November scrape, assembly of notebook, report and slides
- **Aitbek Kaiyrbekuly** location track, text track, trust layer, error analysis, slides
- **Both** selection rule, readings, conclusions

- **AI tools** an AI coding assistant (Claude) helped write and debug the code of krisha_ml, the scripts and the app, review the Endterm against the rubric, draft the slides and report tables from our outputs, and check our English
- **Ours** every experiment run and checked by us; the methods, the selection rule, the one-time test, the readings and the conclusions are our own

Conclusions

- **Endterm reproduced, CV honest** duplicates few (+0.015 M), repaired folds are the primary yardstick
- **New sources carry signal** best single blocks −0.15 to −0.24 M, none passes 0.354 M alone
- **Together they pass** network + base blend, CV 6.100 M (base 6.723)
- **Test, once** MAE 5.80 M, MdAPE 7.82 % (Endterm 6.29 M, 9.13 %)
- **Main gain for the user** calibrated interval + reasons + 5 real comparables + adaptive flag, in an app
- **Next** follow ads over time (price cuts, removal) → time-to-sale data, then decision models

Backup: location features, permutation importance



Backup: description flags against the base model's error

